

Readings memo: equivalence scales and sequential fertility

Fertility–housing project

July 18, 2026 — readings only; no model change is proposed here

Why these readings. The model’s income risk is capped by feasibility: every household must afford $\bar{c}(n) + r\bar{h}(n)$ each period, the childless bundle is roughly a third of mean income, and no measured safety-net guarantee for childless adults comes close to that. The July 18 frontier computation makes the bind exact: at the calibrated persistence the model is infeasible above $\sigma_z \approx 0.09$ without intervention; the measured floor buys no room at the current intercept; lowering $\bar{c}_0$ to 0.10 of mean income plus the floor reaches $\sigma_z = 0.20$, where the estate tail nearly closes but young liquid wealth overshoots its target seven-fold. Two literatures speak to the two structural exits. Equivalence scales replace subtraction with division in preferences, so no income realization is ever infeasible. Sequential-fertility models make the *timing* of births a priced choice, which changes how fertility absorbs risk and housing costs and which moments identify it.

1. Equivalence scales. An equivalence scale is a divisor $m(n)$ that converts household spending into a per-adult-equivalent standard of living. The expert scales are conventions: the OECD “Oxford” scale assigns 1 to the first adult, 0.7 to further adults, 0.5 to a child; the OECD-modified scale (Hagenaars, de Vos and Zaidi 1994) uses 0.5 and 0.3; the square-root scale divides by $\sqrt{\text{household size}}$. The measurement tradition estimates m from behavior: Engel’s method reads adult welfare off the food budget share, Rothbarth’s off spending on adult goods, and the two disagree wildly — Deaton and Muellbauer (1986) report that under the most extreme assumptions Engel makes children about four times as expensive as Rothbarth. Barten (1964) scales each commodity separately, so children change relative shadow prices, not just a single deflator. The identification limit is the discipline’s oldest warning: Pollak and Wales (1979) showed that welfare-relevant scale *levels* are not identified from demand data alone, and Blundell and Lewbel (1991) sharpened this to “only changes, not levels.” What survives is either an untestable restriction — Lewbel’s (1989) base independence, under which the scale is a utility-free function of demographics recoverable from shape-invariant demands — or a change of question: Browning, Chiappori and Lewbel (2013) estimate “indifference scales” inside a collective model and find singles need one half to three quarters of a couple’s outlay, couples saving about a third through shared consumption. Quantitative life-cycle work sidesteps estimation and imposes a scale: Fernández-Villaverde and Krueger (2007) deflate consumption by the mean of six published scales; Attanasio, Banks, Meghir and Weber (1999) put demographics in the Euler equation as taste shifters; and — the bridge case for us — Scholz, Seshadri and Khitatrakun (2006) use $n_j U(c_j/n_j)$ with $n_j = (A_j + 0.7K_j)^{0.7}$ and scale their means-tested floor by the same n_j , i.e. scale-in-utility and floor-in-institutions coexist in a published JPE quantitative model.

For our problem the arithmetic is the point. With $u(c/m(n))$ and no translation term, utility is defined for every $c > 0$: however low the income draw, a feasible choice exists, the empty-budget-set region vanishes identically, and σ_z becomes an estimable parameter rather than a feasibility bound. The honest cost is that our nonhomotheticity lives in the intercepts: committed child expenditures make children and housing necessities, which is where the fertility–income and tenure gradients come from. Under pure scaling, child costs turn proportional to the standard of living — a Becker–Lewis flavored substitution, but a different one, and the whole fertility block re-identifies. A modern middle path exists: nonhomothetic CES (Comin, Lashkari and Mestieri 2021) has *no* subtraction term yet generates non-vanishing necessity/luxury structure (their relative Engel elasticities are constants, where Stone–Geary income effects “level off quickly”). Two negative results from this pass: we found no structural model with a housing-specific equivalence scale, and no paper stating the no-empty-budget-set property explicitly — the property is elementary, but we would be asserting it ourselves.

2. Sequential fertility. The dynamic-programming lineage treats fertility as a repeated discrete choice, not a completed quantity. Wolpin (1984) is the canonical estimated model: a per-period birth decision from age 15 to 45 under child mortality. Hotz and Miller (1988) add imperfect conception control, with child-care costs shaping spacing; Arroyo and Zhang (1997) survey the class. Adda, Dustmann and Stevens (2017) put sequential births inside occupation, labor supply and savings choices and price the career cost of children at 35.3% of

lifetime income, three quarters of it from employment interruptions. Two papers sit closest to us. Sommer (2016) embeds a binary per-period birth choice in a Bewley model where infertility risk rises exponentially with age (reaching one at 45) and finds that realistic increases in uninsurable earnings risk postpone births and reduce their number — the tempo and quantum margins moving together. Ejrnæs and Jørgensen (2020) estimate sequential contraceptive effort with income risk and find abortion access functions as an insurance margin. Doepke and Kindermann (2019) make each birth require both partners’ agreement, a veto structure that matters for policy targeting; Baudin, de la Croix and Gobbi (2015) split childlessness into poverty-driven (2.5% of women) and opportunity-cost-driven (8.1%) components. De la Croix and Pommeret (2021) make the option value explicit: postponing a birth is delaying an irreversible investment, and enough uncertainty pushes first births past a threshold — against an age-declining fecundity schedule calibrated to demographic data. The demographers’ vocabulary is Bongaarts and Feeney (1998): quantum is the lifetime number, tempo the mean age, and observed TFR confounds them. The reduced-form housing evidence is timing-shaped too: Dettling and Kearney (2014) find house-price increases raise owner births (+5% per \$10,000) and depress renter births (−2.4%); Lovenheim and Mumford (2013) find \$100,000 of housing wealth raises the probability of a birth by 16–18% with no renter effect.

Read against our architecture: we already have period-by-period births in a fecund window with the (n, s) child states, so “sequential” is not the missing piece — *pricing time* is. Without age-declining fecundity, waiting is costless, so the option value of postponement is mispriced and risk responses load entirely on the number of children (our $\sigma_z = 0.20$ probe raises childlessness from 0.19 to 0.28 at fixed parameters); with a biological clock, part of that response becomes delayed first births and compressed spacing, which is what the data’s timing moments measure. A fecundity schedule is two externally calibrated parameters, and it converts unused moments — age at first birth, spacing, birth hazards by age and by number of children already born — into identifying variation. It would also let the second-birth housing-response target be what its name says, rather than housing-demand discipline.

What this buys, jointly. The two readings answer different halves of one problem. Scales (or nonhomothetic CES) remove the feasibility cap on σ_z at the cost of re-identifying the fertility block; a fecundity-priced sequential structure supplies precisely the new timing moments that block would need, and gives risk a tempo margin our current model lacks. They are complements, each separable, and both are redesigns — nothing here modifies the model today. The near-term income-process question (tail versus young liquid wealth) stands unchanged in the meantime.

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